Assessment Menu: Teacher Guide

The Function Bistro · design notes, grading and answer key

Why students cannot just pick the easy ones

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

Skill group	What it covers	Dishes
NOTATION	Function notation, domain, range and piecewise functions	A1, A2, E1, E2, S2
TRANSFORM	Parent functions and transformations of graphs	A4, A5, E3, E4, S1
COMPOSE	Operations on functions and composition	A6, E5, S3
INVERSE	Inverse functions	A3, E6, S4

If a student only does…	Maximum
Only Appetizers (4 × 5)	20
Appetizers + Entrées (4 × 5 + 4 × 10)	60
Full meal (adds 2 Chef's Specials)	100

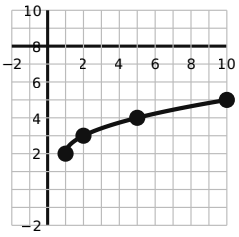
Grading and options

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

Student	Appetizers (of 20)	Entrées (of 40)	Specials (of 40)	Skills OK?	Total

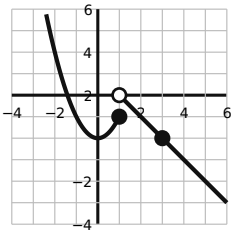
Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Answer graphs are drawn from the functions themselves.

Dish	Problem	Solution	Rubric
A1 NOTATION	Let $f(x) = 2x^2 - 3x + 1$. Find **(a)** $f(-2)$ **(b)** $f(3)$ **(c)** every x for which $f(x) = 1$.	**Answer:** (a) 15 (b) 10 (c) $x = 0$ or $x = \frac{3}{2}$ (a) $2(-2)^2 - 3(-2) + 1 = 8 + 6 + 1 =$ **15** (square −2 first, using parentheses) (b) $2(3)^2 - 3(3) + 1 = 18 - 9 + 1 =$ **10** (c) $2x^2 - 3x + 1 = 1$, so $2x^2 - 3x = 0$ and $x(2x - 3) = 0$, giving **$x = 0$** or **$x = \frac{3}{2}$**.	1 pts: (a) 15 (squares −2 correctly) 1 pts: (b) 10 1 pts: (c) Sets $2x^2 - 3x = 0$ 2 pts: (c) Both solutions: 0 and $\frac{3}{2}$
A2 NOTATION	Relation R = {(1, 3), (2, 5), (3, 3), (4, 7)} and relation T = {(2, 1), (2, 4), (3, 6)}. **(a)** Which one is a function? Explain. **(b)** State the domain and range of that function.	**Answer:** (a) R is a function; T is not (input 2 has two outputs) (b) domain {1, 2, 3, 4}, range {3, 5, 7} (a) In R every input has exactly one output (the output 3 may repeat). In T the input 2 is paired with both 1 and 4, so T is **not** a function. (b) Domain = inputs: {1, 2, 3, 4}. Range = outputs, listing 3 once: {3, 5, 7}.	1 pts: (a) Chooses R 2 pts: (a) Explains: T sends input 2 to two outputs 1 pts: (b) Domain {1, 2, 3, 4} 1 pts: (b) Range {3, 5, 7} (3 listed once)
A3 INVERSE	Let $f(x) = \frac{x - 4}{3}$. **(a)** Find $f^{-1}(x)$. **(b)** Find $f^{-1}(5)$. **(c)** Is $f^{-1}(5)$ equal to $\frac{1}{f(5)}$? Explain.	**Answer:** (a) $f^{-1}(x) = 3x + 4$ (b) 19 (c) No: $\frac{1}{f(5)} = 3$, but $f^{-1}(5) = 19$ (a) Swap x and y: $x = \frac{y - 4}{3}$, so $y - 4 = 3x$ and $y = 3x + 4$. (b) $f^{-1}(5) = 3(5) + 4 =$ **19** (c) The −1 means "inverse," not "reciprocal." $f(5) = \frac{1}{3}$, so $\frac{1}{f(5)} = 3$, which is not 19.	2 pts: (a) Swaps x and y and solves 1 pts: (a) $3x + 4$ 1 pts: (b) 19 1 pts: (c) No: compares 3 and 19
A4 TRANSFORM	The function $g(x) = \lvert x + 3\rvert - 2$ is a transformation of $f(x) = \lvert x\rvert$. **(a)** Describe each shift. **(b)** Give the vertex of $g(x)$. **(c)** Write the equation of the reflection of $g(x)$ in the x-axis.	**Answer:** (a) left 3, down 2 (b) (−3, −2) (c) $y = -\lvert x + 3\rvert + 2$ (a) $x + 3$ moves the graph **left 3**; the −2 moves it **down 2**. (b) The vertex (0, 0) moves to **(−3, −2)**. (c) Reflection in the x-axis is $y = -g(x) = -(\lvert x + 3\rvert - 2) =$ **$-\lvert x + 3\rvert + 2$** (negate both terms).	1 pts: (a) Left 3 (not right) 1 pts: (a) Down 2 1 pts: (b) (−3, −2) 2 pts: (c) $-\lvert x + 3\rvert + 2$ (negates the whole function)
A5 TRANSFORM	Graph $g(x) = \sqrt{x - 1} + 2$ on the grid. Plot the starting point and at least two more points, then draw the curve. State the domain and range.	**Answer:** Starts at (1, 2) and passes through (2, 3) and (5, 4). Domain $x \geq 1$, range $y \geq 2$. The parent $y = \sqrt{x}$ starts at (0, 0). Shift **right 1** and **up 2**: start at (1, 2). Points: $x = 2$ gives 3, $x = 5$ gives 4, $x = 10$ gives 5. Domain: $x - 1 \geq 0$, so **$x \geq 1$**. Range: **$y \geq 2$**. 	1 pts: Starting point (1, 2) 2 pts: Two more correct points and a smooth curve rising to the right 1 pts: Domain $x \geq 1$ 1 pts: Range $y \geq 2$
A6 COMPOSE	Let $f(x) = 2x + 1$ and $g(x) = x^2 - 3$. Find **(a)** $(f(x) + g(x))(2)$ **(b)** $(f \circ g)(2)$ **(c)** $(g \circ f)(2)$.	**Answer:** (a) 6 (b) 3 (c) 22 (a) $f(2) + g(2) = 5 + 1 =$ **6** (b) $(f \circ g)(2) = f(g(2)) = f(1) =$ **3** (do g first) (c) $(g \circ f)(2) = g(f(2)) = g(5) =$ **22**	1 pts: (a) 6 2 pts: (b) Finds $g(2) = 1$ first, then $f(1) = 3$ 2 pts: (c) Finds $f(2) = 5$ first, then $g(5) = 22$

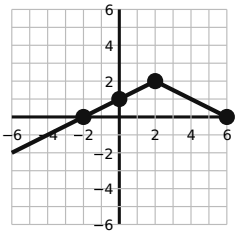
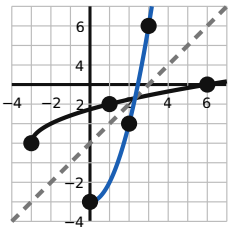
Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Answer graphs are drawn from the functions themselves.

Dish	Problem	Solution	Rubric
E1 NOTATION	Write in interval notation. **(a)** Domain and range of $h(x) = \sqrt{x-5} + 1$. **(b)** Domain and range of $k(x) = \frac{1}{x+2} - 3$. **(c)** Domain of $m(x) = \frac{\sqrt{x+1}}{x-3}$.	**Answer:** (a) $[5, \infty)$; $[1, \infty)$ (b) $(-\infty, -2) \cup (-2, \infty)$; $(-\infty, -3) \cup (-3, \infty)$ (c) $[-1, 3) \cup (3, \infty)$ (a) $x - 5 \geq 0$ gives **[5, ∞)**. The root is ≥ 0, so $h \geq 1$: range **[1, ∞)**. (b) $x + 2 \neq 0$, so the domain is **(−∞, −2) ∪ (−2, ∞)**. The fraction is never 0, so k never equals -3: range **(−∞, −3) ∪ (−3, ∞)**. (c) Need $x + 1 \geq 0$ and $x - 3 \neq 0$: **[−1, 3) ∪ (3, ∞)**.	2 pts: (a) Domain $[5, \infty)$ 1 pts: (a) Range $[1, \infty)$ 2 pts: (b) Domain excludes -2 2 pts: (b) Range excludes -3 (horizontal asymptote) 2 pts: (c) $x \geq -1$ from the square root 1 pts: (c) Also excludes 3, written as a union
E2 NOTATION	Let $f(x) = \begin{cases} x^2 & \text{if } x \leq 1 \\ 3 - x & \text{if } x > 1 \end{cases}$. **(a)** Find $f(-2)$, $f(1)$ and $f(3)$. **(b)** Graph $f(x)$ on the grid using open and closed circles. **(c)** State the range.	**Answer:** (a) 4, 1, 0 (b) see graph: closed circle at (1, 1), open circle at (1, 2) (c) all real numbers (a) $-2 \leq 1$ uses x^2: 4. $x = 1$ uses x^2: 1. $3 > 1$ uses $3 - x$: 0. (b) Parabola piece for $x \leq 1$ ends at a **closed** circle (1, 1). Line piece for $x > 1$ starts at an **open** circle (1, 2) and passes through (3, 0). (c) The parabola piece gives $y \geq 0$ and the line piece gives $y < 2$. Together they cover **all real numbers**. 	1 pts: (a) $f(-2) = 4$ 1 pts: (a) $f(1) = 1$ (uses the first rule) 1 pts: (a) $f(3) = 0$ 2 pts: (b) Parabola piece for $x \leq 1$ 1 pts: (b) Line piece through (3, 0) 1 pts: (b) Closed circle (1, 1) 1 pts: (b) Open circle (1, 2) 2 pts: (c) All real numbers (the two ranges overlap)
E3 TRANSFORM	Start with $f(x) = x^2$. In this order: stretch vertically by a factor of 3, reflect in the x-axis, shift left 2, shift up 5. **(a)** Write the equation of the new function $g(x)$. **(b)** State its vertex and range. **(c)** Find $g(0)$.	**Answer:** (a) $g(x) = -3(x + 2)^2 + 5$ (b) vertex $(-2, 5)$, range $y \leq 5$ (c) -7 (a) Stretch and reflect: $-3x^2$. Shift left 2: $-3(x + 2)^2$. Shift up 5: **$-3(x + 2)^2 + 5$**. (b) The vertex (0, 0) moves to **(−2, 5)**. The parabola opens down, so the range is **$y \leq 5$**. (c) $-3(0 + 2)^2 + 5 = -12 + 5 =$ **-7**	2 pts: (a) Coefficient -3 (stretch and reflection) 2 pts: (a) $(x + 2)^2$ (left, so plus) 1 pts: (a) $+ 5$ 1 pts: (b) Vertex $(-2, 5)$ 2 pts: (b) Range $y \leq 5$ 2 pts: (c) -7
E4 TRANSFORM	The point $(4, -6)$ is on the graph of $y = f(x)$. Name the image of this point on each graph. **(a)** $y = f(x + 3) - 2$ **(b)** $y = -f(x)$ **(c)** $y = f(-x)$ **(d)** $y = 2f(x)$ **(e)** $y = f(\frac{1}{2}x)$	**Answer:** (a) (1, −8) (b) (4, 6) (c) (−4, −6) (d) (4, −12) (e) (8, −6) (a) $x + 3 = 4$ gives $x = 1$ (**left** 3); $y: -6 - 2 = -8$. So $(1, -8)$. (b) $-f$ changes the sign of y: (4, 6). (c) $f(-x)$ changes the sign of x: $(-4, -6)$. (d) Vertical stretch doubles y: $(4, -12)$. (e) $\frac{1}{2}x = 4$ gives $x = 8$ (horizontal stretch): (8, −6).	2 pts: (a) (1, −8): moves left, not right 2 pts: (b) (4, 6) 2 pts: (c) $(-4, -6)$ 2 pts: (d) (4, −12) 2 pts: (e) (8, −6)
E5 COMPOSE	Let $f(x) = x^2 + 1$ and $g(x) = 3x - 2$. **(a)** Find $(f \circ g)(x)$ and $(g \circ f)(x)$, simplified. **(b)** Evaluate $(f \circ g)(-1)$. **(c)** Solve $(g \circ f)(x) = 49$.	**Answer:** (a) $9x^2 - 12x + 5$ and $3x^2 + 1$ (b) 26 (c) $x = 4$ or $x = -4$ (a) $(f \circ g)(x) = (3x - 2)^2 + 1 = 9x^2 - 12x + 4 + 1 =$ **$9x^2 - 12x + 5$** $(g \circ f)(x) = 3(x^2 + 1) - 2 =$ **$3x^2 + 1$** (b) $9(-1)^2 - 12(-1) + 5 = 9 + 12 + 5 =$ **26** (c) $3x^2 + 1 = 49$, so $x^2 = 16$ and **$x = 4$ or $x = -4$**.	3 pts: (a) $(f \circ g)(x) = 9x^2 - 12x + 5$ (squares the binomial correctly) 2 pts: (a) $(g \circ f)(x) = 3x^2 + 1$ 2 pts: (b) 26 1 pts: (c) Sets $3x^2 + 1 = 49$ 2 pts: (c) Both solutions, 4 and -4
E6 INVERSE	Let $f(x) = (x - 2)^2 + 1$ for $x \geq 2$. **(a)** Find $f^{-1}(x)$. **(b)** State the domain and range of $f^{-1}(x)$. **(c)** Verify that $f(f^{-1}(x)) = x$. **(d)** Why is the restriction $x \geq 2$ needed?	**Answer:** (a) $f^{-1}(x) = 2 + \sqrt{x - 1}$ (b) domain $[1, \infty)$, range $[2, \infty)$ (c) see steps (d) so f is one-to-one (a) Swap: $x = (y - 2)^2 + 1$, so $y - 2 = \pm\sqrt{x - 1}$. Since $y \geq 2$, take the plus: **$2 + \sqrt{x - 1}$**. (b) The domain of f^{-1} is the range of f: **[1, ∞)**; its range is the domain of f: **[2, ∞)**. (c) $f(f^{-1}(x)) = (\sqrt{x - 1})^2 + 1 = (x - 1) + 1 = x$ ✓ for $x \geq 1$. (d) Without it the parabola fails the horizontal line test (not one-to-one), so no inverse *function* exists.	2 pts: (a) Swaps x and y and solves for y 2 pts: (a) Chooses the + root: $2 + \sqrt{(x - 1)}$ 2 pts: (b) Domain $[1, \infty)$ and range $[2, \infty)$ 2 pts: (c) Composition simplifies to x 2 pts: (d) One-to-one / horizontal line test

Answer Key: Chef's Specials (20 pts each)

Worked solutions and rubrics. Answer graphs are drawn from the functions themselves.

Dish	Problem	Solution	Rubric
S1 TRANSFORM	Graph $g(x) = -\frac{1}{2}\lvert x-2\rvert + 2$ on the grid by transforming $y = \lvert x\rvert$. **(a)** List the four transformations. **(b)** Give the vertex, domain and range. **(c)** Find both intercepts exactly and plot them. **(d)** Is $g(x)$ even, odd or neither? Justify.	**Answer:** (a) right 2, compress, reflect, up 2 (b) (2, 2); all reals; $y \le 2$ (c) (−2, 0), (6, 0), (0, 1) (d) neither (a) Shift right 2; compress by $\frac{1}{2}$; reflect in the x-axis; shift up 2. (b) Vertex **(2, 2)**; domain all reals; opens down, range **$y \le 2$**.  (c) y-int: $g(0) = -\frac{1}{2}(2) + 2 = \mathbf{1}$. x-ints: $\lvert x-2\rvert = 4$, so $x - 2 = \pm 4$ and **$x = 6$ or $x = -2$**. (d) $g(-x) = -\frac{1}{2}\lvert x+2\rvert + 2$ is neither $g(x)$ nor $-g(x)$: **neither**. Graph slopes: $\frac{1}{2}$ left of the vertex, $-\frac{1}{2}$ right.	2 pts: (a) Right 2; vertical compression 2 pts: (a) Reflection in the x-axis; up 2 2 pts: (b) Vertex (2, 2) 2 pts: (b) Domain all reals; range $y \le 2$ 3 pts: (c) x-intercepts (−2, 0) and (6, 0) 2 pts: (c) y-intercept (0, 1) 2 pts: (d) Neither, comparing $g(-x)$ with both $g(x)$ and $-g(x)$ 2 pts: (d) Correct reasoning shown 1 pts: Graph: V opening down, vertex (2, 2) 2 pts: Graph: slopes of $\pm\frac{1}{2}$ through intercepts
S2 NOTATION	A garage charges $F(t)$ dollars to park for t hours: $F(t) = \begin{cases} 5 & \text{if } 0 < t \le 2 \\ 3t-1 & \text{if } 2 < t \le 6 \\ 17 & \text{if } 6 < t \le 12 \end{cases}$ **(a)** Find $F(1.5)$, $F(4)$ and $F(10)$. **(b)** For what t is $F(t) = 14$? **(c)** State the domain and range, and what 17 means. **(d)** Dana parks 6.5 hours and computes $3(6.5) - 1$. Find her error.	**Answer:** (a) \$5, \$11, \$17 (b) t = 5 (c) domain (0, 12], range [5, 17]; the most anyone pays is \$17 (d) 6.5 is in the third piece: the charge is \$17, not \$18.50 (a) 1.5 is in the first piece: 5. 4 is in the second: $3(4) - 1 = 11$. 10 is in the third: 17. (b) Only the second piece can give 14: $3t - 1 = 14$, so **$t = 5$** (5 is between 2 and 6 ✓). (c) Domain **(0, 12]**; outputs run from 5 to 17: range **[5, 17]** (fee capped at \$17). (d) 6.5 > 6 uses the third rule: \$17. Her 18.50 used the wrong piece.	2 pts: (a) F(1.5) = 5 2 pts: (a) F(4) = 11 2 pts: (a) F(10) = 17 1 pts: (b) Uses the middle piece 2 pts: (b) t = 5 2 pts: (c) Domain (0, 12] 2 pts: (c) Range [5, 17] 1 pts: (c) 17 is the daily maximum 3 pts: (d) Says 6.5 belongs to the third piece (t > 6) 3 pts: (d) Correct charge \$17
S3 COMPOSE	Use the table. **(a)** Find $(f \circ g)(1)$. **(b)** Find $(g \circ f)(2)$. **(c)** Solve $(f \circ g)(x) = 0$. **(d)** Let $p(x) = \sqrt{x}$ and $q(x) = x - 4$. Find $(p \circ q)(x)$ and its domain.	**Answer:** (a) 5 (b) −1 (c) $x = -1$ (d) $\sqrt{x-4}$, domain $[4, \infty)$ (a) $g(1) = 3$ and $f(3) = 5$, so **5**. (b) $f(2) = 0$ and $g(0) = -1$, so **−1**. (c) $f(u) = 0$ only when $u = 2$. Need $g(x) = 2$: $g(-1) = 2$, so **$x = -1$**. (d) $(p \circ q)(x) = p(x-4) = $ **$\sqrt{x-4}$**. Need $x - 4 \ge 0$: domain **$[4, \infty)$**.	4 pts: (a) g(1) = 3 (inside function first), then f(3) = 5 4 pts: (b) f(2) = 0, then g(0) = −1 6 pts: (c) f(u) = 0 at u = 2 (2); g(x) = 2 (2); x = −1 (2) 3 pts: (d) $\sqrt{x-4}$ (q goes inside p) 3 pts: (d) Domain [4, ∞)
S4 INVERSE	Let $f(x) = \sqrt{x+3}$. **(a)** Graph $f(x)$ and the line $y = x$ on the grid. **(b)** Graph $f^{-1}(x)$ by reflecting $f(x)$ in $y = x$. **(c)** Find $f^{-1}(x)$ algebraically. **(d)** State the domain and range of $f(x)$ and $f^{-1}(x)$. **(e)** Does $y = x^2$ (all real x) have an inverse function? Explain.	**Answer:** (c) $f^{-1}(x) = x^2 - 3$ for $x \ge 0$ (e) No: not one-to-one (a) Points on f: (−3, 0), (−2, 1), (1, 2), (6, 3). (b) Swap coordinates: (0, −3), (1, −2), (2, 1), (3, 6); the curves mirror across $y = x$.  (c) $x = \sqrt{y+3}$, so $x^2 = y + 3$ and $y = x^2 - 3$, with **$x \ge 0$** (the range of f). (d) f: domain $[-3, \infty)$, range $[0, \infty)$; f^{-1}: domain $[0, \infty)$, range $[-3, \infty)$. (e) No: (−2, 4) and (2, 4) share an output, so it fails the horizontal line test.	3 pts: (a) f and y = x graphed, f starting at (−3, 0) 4 pts: (b) Inverse graph with coordinates swapped, starting at (0, −3) 5 pts: (c) Swaps x and y (1), isolates $x^2 - 3$ (2), states $x \ge 0$ (2) 4 pts: (d) Domain and range of f and f^{-1} (swapped) 4 pts: (e) No, with a reason such as (−2, 4) and (2, 4)

x	−1	0	1	2	3
$f(x)$	4	1	−2	0	5
$g(x)$	2	−1	3	1	0